

Linux **page_owner**: memory profiling at page granularity with stack traces

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Overview

-  Where is memory going?
-  What is page_owner?
-  What is it for?
-  How it works?
-  How to use it?
-  Basics
-  Examples



Where is memory going?

- 🔍 How do you identify where memory is going in a system?
- 🤔 Statistics might help... or not... and can be confusing 🤪.

```
$ free -g
              total        used  ...
Mem:           54          21  ...
Swap:          47           0  ...
```

```
$ ps -eo '%mem,pid,command' \
  --sort '-%mem' --width 40
%MEM    PID COMMAND
14.8 1495438 qemu-system-x86_64 -accel k
2.6  81749  /usr/lib/firefox-esr/firefo
1.5  82370  /usr/lib/firefox-esr/firefo
1.4 1498905 /usr/lib/firefox-esr/firefo
1.2 1506737 /usr/lib/firefox-esr/firefo
1.0 1251817 /usr/lib/firefox-esr/firefo
1.0  80830  /usr/bin/gnome-shell
0.9  82268  /usr/lib/firefox-esr/firefo
0.8 1503956 /usr/lib/libreoffice/progra
...
```

```
$ cat /proc/meminfo \
  | numfmt --field=2 --grouping
...
Buffers:           1,119,088 kB
Cached:            30,654,480 kB
SwapCached:         284 kB
...
Active(anon):       16,468,212 kB
Inactive(anon):      226,752 kB
Active(file):        10,044,232 kB
Inactive(file):      21,349,864 kB
...
AnonPages:          16,263,256 kB
...
KReclaimable:        2,485,656 kB
Slab:                 2,990,108 kB
SReclaimable:         2,485,656 kB
SUnreclaim:          504,452 kB
...
```



Where is memory going?

- 🔍 What if you could find... what is consuming
 - **each page of memory** (4KiB on x86_64) in the system
 - along with the **process name** and PID
 - and kernel **stack trace**?



- ✅ You can do that with **page_owner**!

What is page_owner?

-  It is a **debug feature** for memory allocation/usage (with high overhead **!**).
-  It stores the **kernel stack trace** (and more) of memory allocation **for every page**.
-  For example: this **one** page, allocated by **dd**, doing a **file write**, to tmpfs (**shmem**):

```
# cat /sys/kernel/debug/page_owner
...

Page allocated via order 0, mask 0x...(GFP_HIGHUSER_MOVABLE|__GFP_COMP), pid 228, tgid 228 (dd), ts 8716052197 ns
PFN 0xccf9 type Movable Block 102 type Movable Flags 0x...(referenced|uptodate|dirty|lru|swapbacked|node=0|zone=1)
  get_page_from_freelist+0x13b9/0x1530
  ...
  shmem_alloc_folio+0x97/0xb0
  ...
  shmem_file_write_iter+0x81/0x90
  vfs_write+0x28d/0x450
  ...
  entry_SYSCALL_64_after_hwframe+0x77/0x7f
Charged to memcg session-1.scope

...
```

What is it for?

-  It provides a **system-wide snapshot/sample** of memory allocations
 - Per **page**
 - Per **stack trace**
-  It can answer the questions:
 - **What** stack trace allocated **this** page? (its “owner”)
 - **How many** pages are allocated by **this** stack trace?

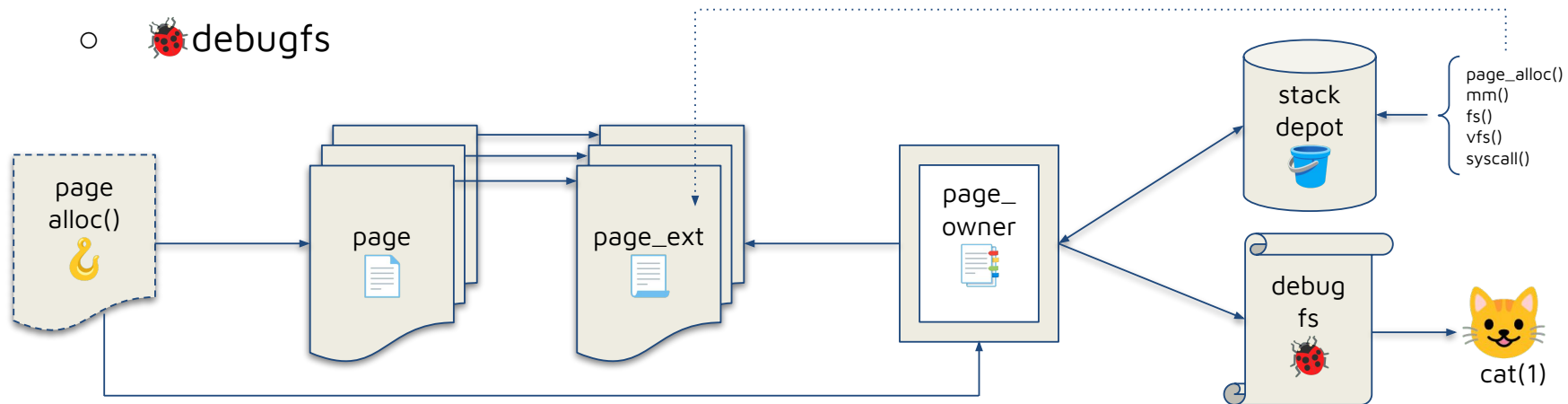
What is it for?

-  It can be used for:
 -  **Sampling**: detail the *system-wide* memory consumption *at a point in time*
 - Examples: analyze memory usage; identify top memory consumers
 -  **Profiling**: understand the *per-process/stack trace* memory consumption *over time*
 - How many pages are used by each process and/or by each stack trace?
 - Examples: analyze memory usage of particular process(es)/stack trace(s)
 -  **Monitoring**: watch a *particular process/stack trace's* memory consumption *over time*
 - How many pages are used by some memory allocation stack trace(s)?
 - Examples: testing code changes; verify memory-constrained cases



How it works?

-  page_owner combines:
 -  page allocation hooks
 -  page_ext (*extension*) for struct page 
 -  stack_depot
 -  debugfs



How to use it?

- Preparation:
 1. 🏗️ Build the kernel with **CONFIG_PAGE_OWNER=y** (mm/Kconfig.debug)
 2. 🥾 Boot the kernel with **page_owner=on** (kernel cmdline)
- Stack traces (and more!) per page:
 - 🐱 cat /sys/kernel/debug/**page_owner**
- Pages per stack trace:
 - 🐱 cat /sys/kernel/debug/**page_owner_stacks/show_stacks**
- Pages per stack trace “handle” (identifier) and mapping from handles to stack traces:
 - 🐱 cat /sys/kernel/debug/**page_owner_stacks/show_handles**
 - 🐱 cat /sys/kernel/debug/**page_owner_stacks/show_stacks_handles**

How to use it?



New feature (full output with stack trace “handle” numbers)



Yet to be released in Linux v7.3.

- Stack traces “handle” numbers (an unique identifier) and more!, per page:
 - (example with file descriptor number 3)
 - `exec 3<>/sys/kernel/debug/page_owner`
 - `echo 'mode=handle' >&3`
 -  `cat <&3`
 - `exec 3>&-`



Basics (0/4)



Write a 256 MiB file to tmpfs.

```
# mount -t tmpfs -o size=1G none /tmpfs  
# dd if=/dev/zero of=/tmpfs/file bs=1M count=256
```



Basics (1/4)



What stack trace allocated this page? (and more!)

```
# cat /sys/kernel/debug/page_owner # (stacks per page)
```

```
...
```

```
Page allocated via order 0, mask 0x...(GFP_HIGHUSER_MOVABLE|__GFP_COMP), pid 228, tgid 228 (dd), ts 8716052197 ns  
PFN 0xccf9 type Movable Block 102 type Movable Flags 0x...(referenced|uptodate|dirty|lru|swapbacked|node=0|zone=1)  
get_page_from_freelist+0x13b9/0x1530
```

```
...
```

```
shmem_alloc_folio+0x97/0xb0
```

```
...
```

```
shmem_file_write_iter+0x81/0x90
```

```
vfs_write+0x28d/0x450
```

```
...
```

```
entry_SYSCALL_64_after_hwframe+0x77/0x7f
```

```
Charged to memcg session-1.scope
```

```
# cat /sys/kernel/debug/page_owner | grep -c '^Page allocated via order 0, .* (dd)' # (zero-order pages allocated by dd)  
65553
```

```
# echo $((65553*4096/1024**2)) # (number of 4 KiB pages in MiB; approximately 256 MiB)  
256
```



Basics (2/4)



How many pages are allocated by this stack trace?

```
# cat /sys/kernel/debug/page_owner_stacks/show_stacks # (pages per stack)
...

get_page_from_freelist+0x13b9/0x1530
...
shmem_alloc_folio+0x97/0xb0
...
shmem_file_write_iter+0x81/0x90
vfs_write+0x28d/0x450
...
entry_SYSCALL_64_after_hwframe+0x77/0x7f
nr_base_pages: 65649
...

# echo $((65649*4096/1024**2)) # (number of 4 KiB pages in MiB; approximately 256 MiB)
256
```



Basics (3/4)



How many pages are allocated by this stack trace handle?

```
# cat /sys/kernel/debug/page_owner_stacks/show_handles
...
handle: 23199752
nr_base_pages: 65649
...

# cat /sys/kernel/debug/page_owner_stacks/show_stacks_handles
...
get_page_from_freelist+0x13b9/0x1530
...
shmem_alloc_folio+0x97/0xb0
...
shmem_file_write_iter+0x81/0x90
vfs_write+0x28d/0x450
...
entry_SYSCALL_64_after_hwframe+0x77/0x7f
handle: 23199752
...
```



Basics (4/4)



What stack trace “handle” allocated this page? (and more!)  for Linux v7.3

```
# exec 3<>/sys/kernel/debug/page_owner # (open in read-write mode as file descriptor 3)
```

```
# echo 'mode=handle' >&3 # (write to fd 3: set print mode to handles-only, no stack traces)
```

```
# cat <&3 # (read from fd 3)
```

```
...
```

```
Page allocated via order 0, mask 0x...(GFP_HIGHUSER_MOVABLE|__GFP_COMP), pid 228, tgid 228 (dd), ts 8716052197 ns  
PFN 0xccf9 type Movable Block 102 type Movable Flags 0x...(referenced|uptodate|dirty|lru|swapbacked|node=0|zone=1)
```

```
handle: 23199752
```

```
Charged to memcg session-1.scope
```

```
# exec 3>&1 # (close fd 3)
```



Examples (1/3)



Sampling: detail the system-wide memory consumption at a point in time



Examples (1/3)



Sampling: detail the system-wide memory consumption at a point in time

Right after boot:

```
# cat /sys/kernel/debug/page_owner >po.boot
# cat /sys/kernel/debug/page_owner_stacks/show_stacks >po.boot.stacks
# cat /sys/kernel/debug/page_owner_stacks/show_handles >po.boot.handles
# cat /sys/kernel/debug/page_owner_stacks/show_stacks_handles >po.boot.stacks_handles

# exec 3<>/sys/kernel/debug/page_owner
# echo 'mode=handle' >&3
# cat <&3 >po.boot.mode-handle
# exec 3>&-

# du -h po.boot* | sort -rh
350M    po.boot
199M    po.boot.mode-handle
1.1M    po.boot.stacks_handles
936K    po.boot.stacks
64K     po.boot.handles
```



Examples (1/3)



Sampling: detail the system-wide memory consumption at a point in time

Top memory consuming processes: (differently than `top`, this includes earlier processes and more kernel memory)

Example line:

Page allocated via order 0, ..., pid 894, tgid 894 ((udev-worker)), ...

Helper:

```
memory_used_per_command()
{
    local FILE="$1"

    cat "$FILE" \
    | awk -F '[ ,]' \
        '/^Page allocated/ { ORDER=$5; COMM=$15; PAGES[COMM]+=2^ORDER }
        END { for (COMM in PAGES) { print COMM, (PAGES[COMM] * 4096 / 1024 ^ 2) } }' \
    | sort -rV -k2,2 \
    | column -t -N 'COMMAND,MiB' \
    | sed 's/(//; s/)//;'
}
```



Examples (1/3)



Sampling: detail the system-wide memory consumption at a point in time

Top memory consuming processes:

Usage:

```
# memory_used_per_command po.boot.mode-handle | head
```

COMMAND	MiB
swapper/0	1086.24
cat	486.141
gnome-shell	226.883
(udev-worker)	212.422
PK-Backend	189.535
khugepaged	105.746
ext4lazyinit	65.7305
gs-plugin-appst	64.3555
alacritty	57.7383

Drill down on the top 3 processes by stack trace...



Examples (1/3)



Sampling: detail the system-wide memory consumption at a point in time

Drill down on the top 3 processes by stack trace:

Example lines:

```
Page allocated via order 0, ..., pid 894, tgid 894 ((udev-worker)), ...  
handle: 387284
```

Helper:

```
memory_used_per_command_handle()
```

```
{  
    local FILE="$1"  
    local COMM="$2"  
  
    cat "$FILE" \  
    | perl -00ne "print if /\($COMM\)/" \  
    | awk \  
        '/^Page allocated/ { ORDER=$5 }  
        /^handle:/ { HANDLE=$2; PAGES[HANDLE]+=2^ORDER }  
        END { for (HANDLE in PAGES) { print HANDLE, (PAGES[HANDLE] * 4096 / 1024 ^ 2) } }' \  
    | sort -rV -k2,2 \  
    | column -t -N 'HANDLE,MiB'  
}
```



Examples (1/3)



Sampling: detail the system-wide memory consumption at a point in time

Drill down on the top 3 processes by stack trace: (cat)

Usage:

```
# memory_used_per_command_handle po.boot.mode-handle 'cat' | head
```

HANDLE	MiB
56754202	472.145
87293984	11.2656
40501283	1.14062
90964000	1.0625
27787272	0.0742188
32899080	0.00390625
30539784	0.00390625
132775944	0.00390625
131596296	0.00390625

Check the top memory consuming stack trace handle for this command.



Examples (1/3)



Sampling: detail the system-wide memory consumption at a point in time

Drill down on the top 3 processes by stack trace:

Example lines:

```
get_page_from_freelist+0x17ff/0x1af0
__alloc_frozen_pages_noprof+0x17c/0x310
...
ksys_write+0x6b/0xe0
do_syscall_64+0xea/0x640
entry_SYSCALL_64_after_hwframe+0x76/0x7e
handle: 56754202
```

Helper:

```
stack_trace_for_handle()
{
    local FILE="$1"
    local HANDLE="$2"

    cat "$FILE" \
    | perl -00ne "print if /handle: $HANDLE\n/"
}
```



Examples (1/3)



Sampling: detail the system-wide memory consumption at a point in time

Drill down on the top 3 processes by stack trace:

```
# stack_trace_for_handle po.boot.stacks_handles 56754202
get_page_from_freelist+0x17ff/0x1af0
__alloc_frozen_pages_noprof+0x17c/0x310
alloc_pages_mpol+0xa6/0x160
folio_alloc_noprof+0x5b/0xb0
__filemap_get_folio_mpol+0x1f3/0x3f0
ext4_da_write_begin+0x140/0x310 [ext4]
generic_perform_write+0x8b/0x2a0
ext4_buffered_write_iter+0x53/0x100 [ext4]
vfs_write+0x25d/0x440
ksys_write+0x6b/0xe0
do_syscall_64+0xea/0x640
entry_SYSCALL_64_after_hwframe+0x76/0x7e
handle: 56754202
```

This stack trace accounts for most of the memory used by the **cat** command.



Examples (1/3)



Sampling: detail the system-wide memory consumption at a point in time

Drill down on the top 3 processes by stack trace: (gnome-shell)

```
# memory_used_per_command_handle po.boot.mode-handle 'gnome-shell' | head
```

HANDLE	MiB
45875219	91.3789
59244575	43.6445
27787272	34.3086
55050259	10.0586
28704776	8.95703
47972371	6.6875
65667091	5.58203
113246226	5.5
50462739	4.16016

Check the top memory consuming stack trace handle for this command.



Examples (1/3)



Sampling: detail the system-wide memory consumption at a point in time

Drill down on the top 3 processes by stack trace: (gnome-shell)

```
# stack_trace_for_handle po.boot.stacks_handles 45875219
get_page_from_freelist+0x17ff/0x1af0
__alloc_frozen_pages_noprof+0x17c/0x310
alloc_pages_mpol+0xa6/0x160
folio_alloc_noprof+0x5b/0xb0
page_cache_ra_order+0x1a6/0x370
filemap_fault+0x7e0/0x12b0
__do_fault+0x33/0x120
do_fault+0x36c/0x530
__handle_mm_fault+0x969/0xf70
handle_mm_fault+0xf5/0x2e0
do_user_addr_fault+0x211/0x670
exc_page_fault+0x80/0x1a0
asm_exc_page_fault+0x26/0x30
handle: 45875219
```

This stack trace accounts for most of the memory used by the **`gnome-shell`** command.



Examples (1/3)



Sampling: detail the system-wide memory consumption at a point in time

Drill down on the top 3 processes by stack trace: (swapper/0)

```
# memory_by_command_handle po.boot.mode-handle 'swapper\0' | head
```

HANDLE	MiB
1048577	1024.48
28835845	8.15625
38010885	6.28125
88866817	4
86507521	4
84148225	4
22937602	4
126746628	4
75104257	2

Check the top memory consuming stack trace handle for this command.

Examples (1/3)



Sampling: detail the system-wide memory consumption at a point in time

Drill down on the top 3 processes by stack trace: (swapper/0)

```
# stack_trace_for_handle po.boot.stacks_handles 1048577
#

# cat po.boot.mode-handle | perl -00ne 'print if /handle: 1048577\n/' | grep -m1 '^PFN'
PFN 0x1000 type Unmovable Block 8 type Unmovable Flags ...

# cat po.boot | perl -00ne 'print if /PFN 0x1000 /'
Page allocated via order 0, mask 0x0(), pid 1, tgid 1 (swapper/0), ts 248929764 ns
PFN 0x1000 type Unmovable Block 8 type Unmovable Flags ...
  register_early_stack+0x32/0x65
  init_page_owner+0x33/0x250
  page_ext_init+0x247/0x260
  page_alloc_init_late+0x37c/0x390
```

This is an early page not listed in `show_stacks_handles` (it was allocated before `page_owner` initialization.)

This stack trace accounts for most of the memory used by the ``swapper/0`` command.



Examples (2/3)

-  **Profiling**: understand the *per-process/stack trace* memory consumption over time

Examples (2/3)

-  **Profiling**: understand the *per-process/stack trace* memory consumption over time

Let's capture data for Firefox ESR in a few stages:

- 0) Start (file: [po.firefox.0-start](#))
- 1) Restore session (file: [po.firefox.1-restore](#))
- 2) Refresh 3 tabs (file: [po.firefox.2-tabs3refresh](#))
- 3) Open Youtube (file: [po.firefox.3-youtube](#))
- 4) Open WebGL Aquarium (file: [po.firefox.4-aquarium](#))

For each stage:

```
# exec 3<>/sys/kernel/debug/page_owner && echo 'mode=handle' >&3
# cat <&3 >$FILE
# exec 3>&1
```

Later:

```
# cat /sys/kernel/debug/page_owner >po.firefox
# cat /sys/kernel/debug/page_owner_stacks/show_stacks >po.firefox.stacks
# cat /sys/kernel/debug/page_owner_stacks/show_handles >po.firefox.handles
# cat /sys/kernel/debug/page_owner_stacks/show_stacks_handles >po.firefox.stacks_handles
```

Examples (2/3)

-  **Profiling**: understand the *per-process/stack trace* memory consumption over time

How much memory is used in each stage?

Check the top 3 consuming stack traces per stage:

```
# memory_used_per_command_handle po.firefox.0-start 'firefox-esr' | head -n4
```

	HANDLE	MiB
(A)	45875219	87.9102
(B)	27787272	64.3828
(C)	28704776	58.0312

```
# memory_used_per_command_handle po.firefox.1-restore 'firefox-esr' | head -n4
```

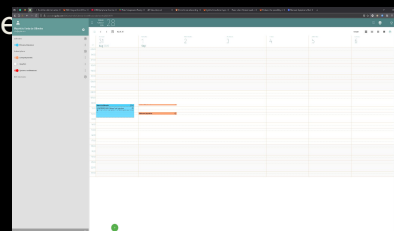
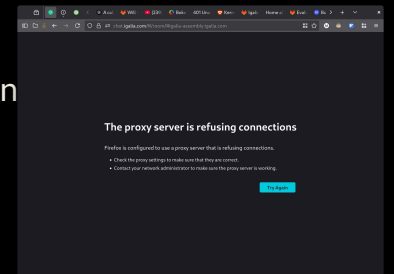
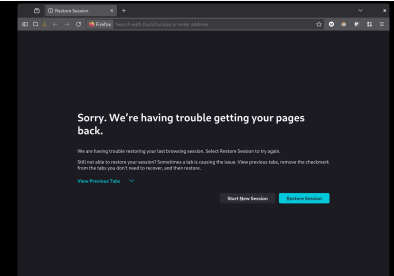
	HANDLE	MiB
(D)	54394900	177.586
(A)	45875219	89.4375
(C)	28704776	70.875

```
# memory_used_per_command_handle po.firefox.2-tabs3refresh 'firefox-esr' | head -n4
```

	HANDLE	MiB
(D)	54394900	177.887
(A)	45875219	110.074
(C)	28704776	77.3594

```
# memory_used_per_command_handle po.firefox.3-youtube 'firefox-esr' | head -n4
```

	HANDLE	MiB
(D)	54394900	177.922



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Examples (2/3)

-  **Profiling**: understand the *per-process/stack trace* memory consumption over time

How much memory is used in each stage?

Check the top 3 consuming stack traces per stage:

(continued)

```
# memory_used_per_command_handle po.firefox.3-youtube 'firefox-esr' | head
```

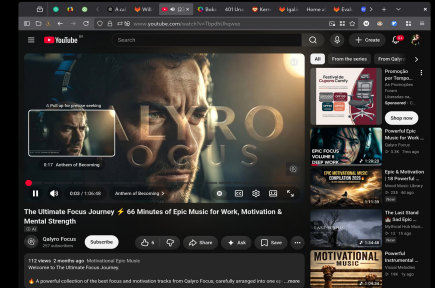
	HANDLE	MiB
(D)	54394900	177.922
(A)	45875219	111.723
(C)	28704776	96.7031

```
# memory_used_per_command_handle po.firefox.4-aquarium 'firefox-esr' | head
```

	HANDLE	MiB
(D)	54394900	243.043
(C)	28704776	109.41
(A)	45875219	98.75

Summary:

(A): 87 -> 89 -> 110 -> 111 -> 98
(B): 64 -> (disappear)
(C): 58 -> 70 -> 77 -> 96 -> 109
(D): 0 -> 177 -> 177 -> 177 -> 243



Examples (2/3)

-  **Profiling**: understand the *per-process/stack trace* memory consumption over time

How much memory is used in each stage?

Check the top 3 consuming stack traces per stage:

(A) read file-backed memory, page fault, page cache, read-ahead

```
# stack_trace_for_handle po.firefox.stacks_handles 45875219
```

```
get_page_from_freelist+0x17ff/0x1af0
```

```
__alloc_frozen_pages_noprof+0x17c/0x310
```

```
alloc_pages_mpol+0xa6/0x160
```

```
folio_alloc_noprof+0x5b/0xb0
```

```
page_cache_ra_order+0x1a6/0x370
```

```
filemap_fault+0x7e0/0x12b0
```

```
__do_fault+0x33/0x120
```

```
do_fault+0x36c/0x530
```

```
__handle_mm_fault+0x969/0xf70
```

```
handle_mm_fault+0xf5/0x2e0
```

```
do_user_addr_fault+0x211/0x670
```

```
exc_page_fault+0x80/0x1a0
```

```
asm_exc_page_fault+0x26/0x30
```

```
handle: 45875219
```

Examples (2/3)

-  **Profiling**: understand the *per-process/stack trace* memory consumption over time

How much memory is used in each stage?

Check the top 3 consuming stack traces per stage:

(C) page fault, mapping memory to userspace (virtual memory area)

```
# stack_trace_for_handle po.firefox.stacks_handles 28704776
get_page_from_freelist+0x17ff/0x1af0
__alloc_frozen_pages_noprof+0x17c/0x310
alloc_pages_mpol+0xa6/0x160
vma_alloc_folio_noprof+0x6e/0xd0
do_fault+0x7d/0x530
__handle_mm_fault+0x969/0xf70
handle_mm_fault+0xf5/0x2e0
do_user_addr_fault+0x211/0x670
exc_page_fault+0x80/0x1a0
asm_exc_page_fault+0x26/0x30
handle: 28704776
```

Examples (2/3)

-  **Profiling**: understand the *per-process/stack trace* memory consumption over time

How much memory is used in each stage?

Check the top 3 consuming stack traces per stage:

(D) page fault, shared memory backing

```
# stack_trace_for_handle po.firefox.stacks_handles 54394900
get_page_from_freelist+0x17ff/0x1af0
__alloc_frozen_pages_noprof+0x17c/0x310
alloc_pages_mpol+0xa6/0x160
folio_alloc_mpol_noprof+0x14/0x40
shmem_alloc_folio+0x9c/0xb0
shmem_alloc_and_add_folio+0x1a7/0x480
shmem_get_folio_gfp+0x20a/0x5e0
shmem_fault+0x82/0x310
__do_fault+0x33/0x120
do_fault+0x16f/0x530
__handle_mm_fault+0x969/0xf70
handle_mm_fault+0xf5/0x2e0
do_user_addr_fault+0x211/0x670
exc_page_fault+0x80/0x1a0
asm_exc_page_fault+0x26/0x30
handle: 54394900
```

Examples (3/3)

-  **Monitoring**: watch a *particular process/stack trace's* memory consumption over time

Examples (3/3)

-  **Monitoring:** watch a *particular process/stack trace's* memory consumption over time

Monitor Firefox ESR's memory when opening tabs.
The (C) handle increases when tabs are opened.

Commands:

```
COMM='firefox-esr'  
HANDLE=28704776 # (C)
```

```
while true; do
```

```
    exec 3<>/sys/kernel/debug/page_owner  
    echo 'mode=handle' >&3  
    cat <&3 >/tmp/po.mode-handle  
    exec 3>&-
```

```
    memory_used_per_command_handle /tmp/po.mode-handle "$COMM" \  
    | head | grep "$HANDLE" | tee /tmp/po.chart
```

```
done
```

Examples (3/3)

-  **Monitoring:** watch a *particular process/stack trace's* memory consumption over time

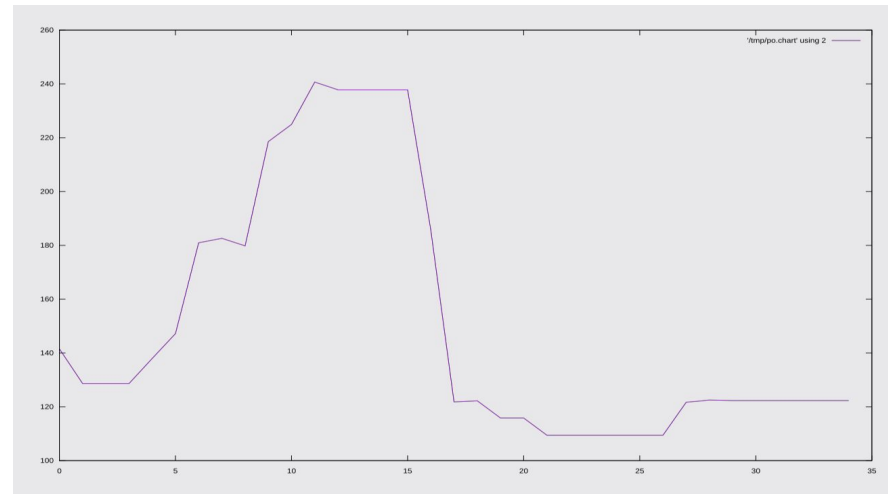
Monitor Firefox ESR's memory when opening tabs.
The (C) handle increases when tabs are opened.

Output:

28704776	141.543
28704776	128.645
28704776	128.645
28704776	128.645
28704776	137.988
28704776	147.191
28704776	180.961
28704776	182.617
28704776	179.809
28704776	218.566
28704776	224.957
28704776	240.688
28704776	237.809
28704776	237.809
28704776	237.809
28704776	237.809
28704776	185.656
28704776	121.809

Gnuplot Command

```
gnuplot -e "set terminal svg size 1280,800;  
set output '/tmp/po.chart.svg';  
set style data line;  
plot '/tmp/po.chart' using 2;"
```





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